

Article

Project-Based Learning in Social Innovation—Developing a Sense of Belonging in Online Contexts

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Abstract

This study examines how students perceive factors contributing to their sense of belonging (SoB) in an online project-based master's course in social innovation, incorporating collaboration with external stakeholders and structured around real-world challenges from external partners. SoB has been shown to impact academic outcomes positively, but online teamwork and collaboration outside the institution walls present unique challenges. The study adopts a qualitative approach and an interpretivist standpoint to find meaningful aspects and generate an understanding of positions among the students. Data was gathered through interviews, allowing students to express their unique experiences and perspectives on SoB in the context of a master's course, with participants from several countries. Thematic analysis of the interview data identified the following seven categories related to students' perceived belonging: team formation and trust, personal introductions, group continuity, access to teachers, use of informal communication channels, shared tools, and digital fluency. Furthermore, working towards a common challenge or goal seems to promote SoB among team members. The results indicate that SoB was shaped by multiple interrelated factors, with team-based collaboration and structured group dynamics playing a central role. The study contributes to ongoing research on student belonging by identifying specific practices that may support SoB in digitally supported, team-oriented learning environments.



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1. Introduction

In recent years, higher education has increasingly embraced online and hybrid learning formats, offering flexibility, and broader access to diverse student populations. However, this shift has also raised concerns about students' social integration and emotional connection to peers, instructors, and institutions. A growing body of research highlights *sense of belonging* (SoB) as a critical factor in academic persistence, motivation, and engagement (Hunn, 2014; Museus et al., 2018; Freeman et al., 2007). The challenges of fostering SoB are magnified in online or digitally mediated learning environments, where opportunities for informal interaction and social bonding are limited.

While much of the existing literature focuses on traditional classroom or campus settings, some recent studies have begun to examine SoB in virtual or blended contexts

(Ahn & Davis, 2020a; Brodie & Osowska, 2021; Peacock & Cowan, 2019). These studies suggest that students' experiences of belonging are shaped not only by content delivery but also by digital interaction tools, group dynamics, and institutional culture. However, there is limited qualitative research that explores how SoB is constructed in online project-based courses that incorporate real-world societal challenges and external stakeholder interaction—an increasingly relevant format in higher education.

This study addresses that gap by investigating SoB among students enrolled in a master-level course in social innovation at Openlab Stockholm. The course was delivered entirely online between 2020 and 2022 due to the COVID-19 pandemic and is characterised by transdisciplinary teamwork, design thinking, and challenge-based learning. Students were required to collaborate in diverse groups and interact with representatives from public-sector organisations.

This study examines how students in a master's course given by Openlab Stockholm describe what contributes to a SoB. The course has been given both on campus (2013–2019 and 2023–2025) and online (2020–2022). This yielded a solid basis for comparisons and included communication with external stakeholders, such as policymakers, employers, and community members. The course and its structure are further described in (Rosén et al., 2022).

The study is guided by the following research question: *What factors do students identify as contributing to a sense of belonging in an online, project-based master's course involving external stakeholders?* This question is relevant for institutions seeking to design more inclusive and engaging digital learning environments, particularly in courses that rely on teamwork, innovation, or societal engagement.

To explore this question, we conducted semi-structured interviews with eight students, allowing them to share their unique experiences and perspectives. The study adopts a qualitative interpretivist approach aimed at identifying meaningful themes rather than imposing predefined analytical categories. This approach enables us to foreground the voices of students and capture the complex interplay between cognitive, emotional, and social dimensions of learning.

The remainder of the paper is structured as follows: we first review relevant literature on sense of belonging in education, then present the Openlab course context and research method. Following that, we describe the method and report the clustered student opinions. The study does not claim to be comprehensive regarding SoB. However, it highlights what the students have described as contributing to the SoB in the current course, with its unique character and the socio-cultural context of the study. We conclude with implications for online education design and a reflection on limitations.

2. Background

The course has been designed to engage students in a real-world challenge, collaboratively tackled within deliberately diverse teams and guided by an innovation process framework. This context is essential for interpreting our findings on belonging, as many of the factors that students mention, including teamwork, stakeholder contact, and quick-channel communication, were direct consequences of this pedagogical structure. Moving a campus course online creates a need for digital channels for communication and a high demand for self-regulated learning (Wang et al., 2013). Consequently, shared communities where students focus on solving problems together and can benefit from each other more socially are important (Kahiigi et al., 2008). In order to create such shared communities, the concept of belonging is central. At the same time, cognitive dimensions of learning have consistently been recognised as central to acquiring knowledge and skills, and a broader corpus of research is beginning to exhibit the importance of students' emotions in

their learning processes. The social dimension of learning involves interaction with others through participation, communication, and cooperation. It introduces learners to adopting the values and norms of unique social, cultural, practical, or vocational traditions. It also aims at using social interaction as a catalyst for high-quality learning.

Project-based learning (PBL) is a student-centred pedagogical approach that organises learning around sustained inquiry into complex, real-world problems (J. W. Thomas, 2000; Bell, 2010). Rather than acquiring knowledge through lectures, students co-construct understanding through collaborative projects that demand creativity, critical thinking, and communication. PBL has been associated with increased student engagement and deeper learning (Holm, 2011; Strobel & van Barneveld, 2009). In particular, the requirement to work in teams over extended periods fosters peer interdependence and shared ownership of learning outcomes, which are positively linked to the development of SoB (Scogin et al., 2017; Blumenfeld et al., 1991). The Openlab course explicitly applies PBL principles by using authentic challenges from external organisations to structure team-based work, creating the conditions for strong collaborative bonds and meaningful student engagement. In this study, project-based learning (PBL) is not only the delivery format, but a key organising structure that shapes social interaction. The open-ended, collaborative, and externally facing nature of PBL in the Openlab course means that students' belonging is likely influenced by how project roles, team dynamics, and stakeholder engagement are experienced in real time.

2.1. Sense of Belonging

Sense of belonging (SoB) is increasingly recognised as a foundational construct in student retention, motivation, and academic success. Goodenow (1993) describes SoB as “psychological membership,” or the feeling of being accepted, valued, and included in an academic setting. Yuval-Davis (2006) expands this to include emotional safety, respect, and a sense of connection. Ahn and Davis (2020a) identify four domains central to SoB: academic engagement, social interaction, personal identity, and environmental surroundings. SoB is not only predictive of academic outcomes (Freeman et al., 2007; L. Thomas, 2012), but also tied to students' willingness to collaborate, persist through difficulty, and seek help (Vaccaro et al., 2015; Peacock & Cowan, 2019). In online (or hybrid) learning environments, SoB becomes harder to establish, often due to reduced informal interaction and limited visibility of others' presence (Brodie & Osowska, 2021). This challenge has prompted various strategies to build online learning communities, including structured introductions, informal chat spaces, and active social presence from instructors (Kear et al., 2014). However, well-designed team-based learning and sustained peer interaction can help mitigate this loss (Kember et al., 2001; S. M. Ross, 2019).

The boundaries when trying to map SoB are broad. Ahn and Davis (2020a) found four domains of relevance: academic engagement, social engagement, surroundings (living space and geographical and cultural location), and personal space (life satisfaction, life attitudes, identity, and personal interests). They conclude “that policies for student engagement in higher education should reflect all four domains to support the full range of student experiences.” (Ahn & Davis, 2020a, p. 622). The concept of SoB relates to a collection of connected ideas, models, and teaching and learning frameworks. Yuval-Davis (2006) describes SoB as a feeling of ease, safety, and being connected and respected. Goodenow (1993) uses the term “psychological membership” to describe the concept. Other researchers utilise different relationships between SoB and social capital when interpreting the concept (Ahn & Davis, 2020b). Brodie and Osowska (2021, p. 5) state the highly relevant point that “some students did not want a sense of belonging and were just interested in getting their qualification.” Courses are often designed with young full-time students in

mind rather than adults with a lot of work and family obligations (Brodie & Osowska, 2021, p. 353), although many institutions, among them Openlab Stockholm, have a much more diversified student base also when it comes to age and walk of life.

2.2. University Belonging Questionnaire

Recent developments in belonging research have produced validated instruments that offer structured insights into students' self-perceived fit within the university context. The University Belonging Questionnaire (Slaten et al., 2018), developed through a multi-stage psychometric process, captures the following five distinct but interrelated dimensions of belonging: social self, competence, university support, connectedness, and experiences of discrimination. The instrument demonstrated strong reliability and factorial validity, with each subscale contributing uniquely to the overarching construct of university belonging. The UBQ's structure resonates with the present study's conceptualisation of belonging as a multidimensional experience shaped by social, academic, and institutional factors.

However, for reasons stated in Section 3, we chose to be inspired by the UBQ rather than using it completely. Of particular relevance are the UBQ findings on how students' sense of connectedness and perceived institutional support relate to broader motivational and learning outcomes. The connectedness and competence dimensions, for instance, showed strong associations with students' academic confidence and satisfaction. These results support the view that belonging is not simply a social construct but is intricately tied to students' engagement with programme content and their interpretation of the educational environment. The integration of such psychometric evidence strengthens the theoretical foundation of the current study and provides external validation for its focus on how structural and cultural aspects of academic programmes influence students' motivation and perceived learning gains.

2.3. Current Literature

In recent years, a growing body of research has examined how sense of belonging (SoB) operates in online and hybrid higher education environments. The shift toward online and blended learning, accelerated by the COVID-19 pandemic, has prompted questions about whether traditional predictors of belonging—such as face-to-face interaction, campus engagement, and informal peer networks—retain their influence when mediated by digital formats. Studies have found that SoB can be sustained in virtual contexts, but often depends on deliberate pedagogical design, clarity of expectations, and responsiveness from instructors (Deng & Tavares, 2021; Gray & DiLoreto, 2016). In contrast, when students encounter poorly structured or inconsistent online learning environments, feelings of isolation and academic detachment tend to increase (Lowenthal et al., 2020). These findings underscore the growing need to treat SoB not as a fixed trait or personality factor, but as a relational outcome shaped by instructional design and institutional practices.

While some SoB studies have focused on social integration or identity-based belonging, there is increasing recognition that cognitive and structural dimensions, such as understanding one's role in a programme, perceiving the course structure as coherent, and having confidence in one's learning progress, are equally critical, particularly in hybrid and asynchronous formats. For example, Kahu et al. (2020) argue that students' academic belonging emerges from a dynamic interplay between personal motivation and programme-related signals of support, structure, and relevance. Similarly, Peacock et al. (2021) identify pedagogical transparency and timely feedback as essential conditions for nurturing SoB in digital settings. These perspectives closely align with the present study's emphasis on programme structure and study motivation, suggesting that institutional efforts to foster

SoB must extend beyond social connection and address the epistemic and organisational environments in which students learn.

The present study contributes to this evolving literature by examining SoB as a function of students' experienced motivation, perceived learning gains, and their alignment with programme culture and structure. Rather than treating online settings as binary conditions, the study approaches belonging as a situated experience that emerges from the broader study environment, including how goals are communicated, how content is organised, and how students make sense of their academic trajectory. In doing so, the study complements existing SoB research by foregrounding the role of motivational alignment and curricular clarity in shaping students' educational engagement.

2.4. Openlab Stockholm—The Course Setting

Openlab is a challenge-driven innovation arena and community providing transdisciplinary courses for master's and PhD students and professionals, as well as a co-creation space and process support for innovation projects.¹ The founding partners of Openlab are the City of Stockholm, the Region of Stockholm, Karolinska Institutet, the Royal Institute of Technology, Stockholm University, and Södertörn University College. The focus of this article is the master's course. It uses real-life societal challenges originating from its public sector partners, viz. the City of Stockholm and the Region of Stockholm.

Openlab uses a wide range of methods, of which design thinking is the fundamental one: "Design thinking is a method and a mindset that reframes the challenges and places the end-users' needs at the centre of innovation." (Björkman, 2021); therefore, the methods the course participants are expected to utilise differ from those used in many traditional courses. The Openlab master's course is workshop-based, and the primary unit the students work in is a group of five to eight students, called a team at Openlab. The course is a full-semester course (20 weeks) in a two-semester system and is offered in both the autumn and spring semesters. The students are expected to work at least 20 h per week actively, and the course evaluations confirm this level of engagement. The course is only graded pass/fail in order to encourage the students to be creative without feeling that they take any "grading risks" from doing so. The teachers strive for maximum diversity in the teams by using parameters such as home country/cultural background², study/subject background, work experience, and a self-estimate along a divergent/convergent thinking scale. The participants "work in interdisciplinary teams with professionals from all backgrounds, creating opportunity for cross-pollination of minds and interesting discussions." (Björkman, 2021). The teams must interact with the external parties that provide "the challenge" (a hard-solved, even wicked, real-world problem from the challenge givers' workplaces) and decide how to organise the work. Individual skills, engagement, and group dynamics are crucial for the teams' success.

The solutions presented by the students become the object of assessment instead of traditional course hand-in assignments. This setup with real-world problems to solve reduces the possibilities for individual work and forms grounds for the collaboration needed. How well a team functions is highly important, and much emphasis is put on planning and decision-making. The Openlab course requires coordination of times for meetings with the external challenge givers and other course activities. This contrasts with traditional educational practices, which have a set schedule for the whole semester.

Project-based learning (PBL) forms the pedagogical foundation of the Openlab course. In contrast to traditional lecture-based formats, PBL immerses students in extended, collaborative work on open-ended problems that mirror real-world complexity. The assigned stakeholder challenges require teams to explore needs, propose interventions, and communicate findings to external partners. This structure creates frequent opportunities for

social interaction, mutual dependency, and reflection, conditions which are conducive to developing a sense of belonging. The course's emphasis on creativity, interdisciplinarity, and iterative teamwork reinforces the social and emotional dimensions of learning alongside the cognitive. This form of PBL positions students as active contributors to real-world solutions, creating conditions where a sense of belonging is shaped not just by peers or instructors, but by the authenticity and visibility of the work itself.

The sudden transition to online formats prompted institutions to rethink how to foster engagement and connection in digital learning environments. As [Zhao and Watterston \(2021\)](#) argue, post-pandemic education must evolve beyond emergency remote teaching to adopt inclusive, adaptive, and student-centred designs that address both cognitive and emotional aspects of learning.

Because the course is intentionally structured around collaborative, interdisciplinary problem-solving in response to societal challenges, the reported sense of belonging must be understood in this pedagogical context. The design inherently invites shared ownership, stakeholder interaction, and active teamwork, each of which is a potential antecedent to belonging.

3. Method

A qualitative research approach was chosen for this study because it offers the most suitable method for exploring the complex, situated, and meaning-laden nature of students' sense of belonging (SoB) within programme structures. SoB is not only a cognitive judgement but also an affective and experiential phenomenon, often shaped by subtle contextual cues and individual interpretations. Qualitative methods enable the capture of such dimensions through open-ended responses, nuanced reflections, and the articulation of perceived coherence or misalignment between institutional design and student experience. As scholars such as [Merriam and Tisdell \(2016\)](#) have argued, qualitative inquiry is especially valuable when the research goal is to understand how individuals make meaning of their environment and how those meanings inform their engagement, motivation, and perceived outcomes. In this context, the use of qualitative survey data serves not merely as anecdotal illustration but as an analytically rigorous way to surface interpretive patterns grounded in students' lived realities.

Nevertheless, the qualitative design also introduces specific limitations. Subjectivity, while methodologically intended, places constraints on generalisability. The findings reflect how participants perceived their own study environment, which may be shaped by prior experience, disciplinary culture, or personal expectations. By explicitly framing the study as qualitative, the aim is not to produce universal claims but to generate insight into the relational dynamics between programme structure, motivation, and belonging. These insights can then inform the design of further studies, including mixed-method or large-scale quantitative work. Clarifying this rationale allows the limitations of scope and transferability to be understood as deliberate trade-offs, aligned with the epistemological goals of the study.

This study was based on qualitative data collected through eight semi-structured interviews with students enrolled in Openlab's project-based master's course during its online delivery period (2020–2022). The interviews served as the primary data source and were designed to capture students' personal experiences, reflections, and perceptions related to their sense of belonging (SoB) in the course.

3.1. Participant Recruitment

The study was based on eight in-depth, semi-structured interviews with students enrolled in the Openlab master's course during its online implementation in the spring and

autumn semesters of 2020. The students were recruited via purposive sampling to ensure a diversity of academic disciplines, age groups, and levels of prior work experience. Of the 52 students participating in total in the courses, ten were selected to balance the gender, age, geographic background and academic background to reflect the student population. Of these ten, eight completed their interviews.

3.2. Interview Guide

The present study focused on what *students describe* as contributing to SoB. We used a semi-structured interview guide for student interviews to avoid a pre-set range of factors, such as the University Belonging Questionnaire (Slaten et al., 2018) and similar, influencing the students' answers. However, the interviewers utilised information from course evaluations and teacher interviews to facilitate follow-up questions and enhance the in-depth interviews.

The interview guide (see Appendix C) included open-ended questions focusing on team dynamics, digital collaboration, interactions with stakeholders and teachers, and students' emotional and social engagement. The guide further focused on perceptions of inclusion and connection. Example questions included the following: "What helped you feel part of the group or course?", "Were there moments you felt excluded or disconnected?", and "How did your team navigate differences in communication, background, or skill level?". The guide was developed with input from both course faculty and researchers involved in the CIVIS e-Belong project.

Studies investigating students' SoB have typically relied on pre-designed questionnaires to collect data, e.g., the UBQ (Slaten et al., 2018). Rather than being limited by predefined questions and response options, we want to explore each student's unique experience and perspective. The current study setting is a master's course involving societal challenges and stakeholders, and with very heterogeneous student categories attending, making it possible to obtain students' different perspectives on SoB.

Unlike pre-designed questionnaires, which rely on a fixed set of items and predefined response categories, the semi-structured interview format used in this study allowed participants to express their thoughts in their own words and on their own terms. This flexibility made it possible to probe further when students raised unexpected themes or articulated their sense of belonging in culturally nuanced or personally specific ways. Prior questionnaire-based studies have been valuable in establishing generalisable patterns, but they tend to operationalise belonging through a priori constructs, potentially overlooking how students interpret, negotiate, or reject such constructs in specific educational settings. By contrast, our approach enabled us to explore the *meanings students themselves attach* to belonging, especially in a context like Openlab's course where diverse cultural, educational, and professional backgrounds converge. As a result, we were able to identify factors and framings that do not typically emerge in standardised instruments, such as relational belonging mediated through external stakeholder interactions or project-based collaboration with societal impact.

While prior course evaluations and teacher insights informed the interview guide, only the interview transcripts were treated as data in this study. No survey data, observational field notes, or quantitative materials were included in the analysis.

3.3. Interview Setup

Course staff approached participants via email after the course had concluded, and all eight who agreed to participate were interviewed remotely via Zoom. The interviews were conducted at the end of the respective semesters, each lasting between 40 and 80 min. All interviews were audio-recorded with participants' written consent and transcribed

verbatim. The study did not require ethical approval from Stockholm University's internal ethics committee, and all participants were informed about their rights, data confidentiality, and the voluntary nature of their involvement.

Of the eight students, although four were from Sweden, only two preferred to speak Swedish and six spoke English during the interviews. Participants varied in age from early 20s to mid-40s and came from various disciplinary backgrounds, see Appendix A. Several had professional work experience prior to enrolling in the course, while others were completing their master's degrees.

3.4. Interview Coding

Each interview was audio-recorded with the participants' consent. The recordings were transcribed verbatim and served as the basis for the thematic analysis. The interview transcripts were coded using the MaxQDA software. Two researchers were involved in the coding process, Ulf Olsson and Cormac McGrath, both with backgrounds in pedagogy and connected to the Centre for the Advancement of University Teaching at Stockholm University. Both are mentioned in the Acknowledgement section. After multiple readings, an initial coding scheme was developed inductively from the data. To ensure coding consistency, the coders independently applied the scheme to a subset of the transcripts and then met to compare interpretations, resolve discrepancies, and refine category definitions. Although no formal inter-rater reliability coefficient (e.g., Cohen's Kappa) was calculated, since the aim was thematic exploration rather than statistical generalisability, this process served to calibrate understanding and enhance trustworthiness of the coding. The remaining interviews were then divided and coded by both researchers following the agreed structure, with iterative discussions to ensure interpretive alignment. Seven categories were found for further analysis. The categories are presented in a separate table in Appendix B.

4. Results

The themes presented below were derived from recurring patterns in the students' accounts. Each category emerged through iterative coding and was informed by multiple expressions or narratives across interviews. The selected quotes are illustrative but not exhaustive in terms of the data supporting each theme. We have added brief analytic bridges to clarify how each quote reflects or supports its associated category.

The following section presents the students' opinions, and a few illustrative quotes from the interviews were added for each category. The different aspects contributing to a SoB were not ranked due to the qualitative design of the study. An ambition to rank the categories could also be futile, as the preferences vary a lot according to their character, and combinations of categories are brought to the fore. For example, consider the student expressing the following:

"Um, I don't know. Actually, well, I think the challenge is very important because if you're not interested in the challenge, you might lose interest in the course. But, ah, I think the team is very important. Um, ah, if you feel that it's a mixture of those two things because if you feel connected to the challenge you also feel connected to the team."
[Student 06]

This quote describes very well the intertwined relationships between the categories. Another student explained, *"I don't care if the group is happy, I don't care if I make friends, we are never going to meet again"* [Student 01]. The cultural and individual influences, including numerous inputs by the teachers, the digital system used, and expectations, would make ranking unreliable and of little use for the purpose of the study. Next, we present the categories.

4.1. Building Trust Within the Team

"I think it is really important. Especially the parts about resolving conflict. Being many strong personalities in the group." [Student 03]

"Really, it saved us later when one person dropped off. We could point to the agreement and move on." [Student 01]

"We didn't talk much about it after the first week, but just having something written down made me feel like we were all committed." [Student 11]

Several students described how creating a team contract early in the course helped establish a sense of mutual commitment, particularly when resolving conflicts or navigating low participation. For many, the contract served not only as a practical tool but also as a symbolic gesture that the team was serious about collaboration. Others mentioned that the act of openly discussing expectations at the outset made it easier to trust teammates and raise concerns later in the course. Common themes mentioned in this category included: early expectation setting, accountability, conflict resolution, and mutual commitment.

Here, the student reflects on the experience of conflict within a team of strong personalities and implies that the process of resolving those conflicts constructively was central to maintaining functional group collaboration. This aligns with the broader theme of "building trust", as trust is both tested and developed through conflict management, especially in early-stage teamwork.

A "team contract" contributes to developing a social presence and responsibility and can positively affect how well a student team will function, thereby influencing student learning and satisfaction. The purpose of a team contract, formed in the initial stage of a team's collaboration phase, was argued as twofold. First, it contributes to the students' socialisation. They learn about each other's interests, motivation, pre-skills, and personal factors that influence their possibility of joining group activities. Secondly, a contract contains rules about how the work in the group should be conducted and how any issues should be dealt with. The agreement contributes to creating a SoB among team members, although it has no legal functionality. The presence of shared agreements or norms within student teams can contribute meaningfully to a sense of belonging (SoB), particularly by clarifying expectations and fostering mutual accountability. Research shows that clearly defined collaborative frameworks enhance psychological safety and group cohesion, both of which are central to perceived belonging (Baik et al., 2019; L. Thomas et al., 2020). When students co-create or collectively adhere to agreed practices, they are more likely to experience the team as an inclusive and supportive micro-environment within the broader academic structure. One important aspect is how to handle team members who do not contribute to the teamwork satisfactorily. Problems with participants not contributing, not answering mail in due time or failing in other ways can be dealt with according to the contract without any unnecessary and harmful delay.

4.2. Personal Profiles

"It helped a lot that everyone uploaded a photo and short introduction. We didn't feel like strangers in the first team meeting." [Student 12, translated]

"I didn't read all the LMS profiles, but the check-in rounds made a difference. Just hearing people's voices and a few facts made it easier to talk later." [Student 11]

"I preferred not to share much in my profile, but it was nice to get a sense of who the others were." [Student 03]

Students had differing experiences with personal profile features in the digital learning environment. Some appreciated the ability to see names, photos, and short bios within

the course system or during live sessions, describing these as useful for lowering social barriers and creating initial familiarity. Others noted that structured “check-in” activities early in the course helped fulfil the same function, especially for those who were less active in the online platform itself.

Themes associated with this category included the following: digital visibility, personalisation, reduced anonymity, and early social anchoring.

Personal profiles, especially those including photos and introductory facts, were perceived as helpful in creating a baseline familiarity among students. These digital introductions helped mitigate the initial social distance in a fully online environment. As one student explained, the awareness of cultural differences required conscious efforts to bridge gaps in understanding:

“...we were from different cultures, which meant we knew we wouldn't understand each other on all levels.” [Student 02]

In this context, personalised profiles played a role in reducing uncertainty and building interpersonal context in the early stages of collaboration. Photos and personalised profiles facilitate a new course's introductory and socialisation phase. Access to each other's personal information should be easy from different parts of a digital learning system (learning management system). However, personal information should not be accessible to others outside the current course environment, as some students do not feel the need, have privacy concerns, or prefer to communicate in other forums (Kear et al., 2014). Adding URLs in the profile area makes it possible to redirect visitors to other social networking forums, private or work-related, such as LinkedIn. Subject expertise information can include former courses and areas the participants find the most interesting. Depending on the course, students in the Openlab course also participated in check-in exercises 3–4 mornings at the beginning of the semester, either online or on campus. These check-ins consist of telling your name and some facts about yourself. The teachers predetermined the facts to tell. This kind of “ice-breaking activity” enhances the SoB by bringing the students closer to each other by discovering some more personal facts, but the importance was considered differently by the students.

4.3. Teamwork

“And then in our small groups, we kind of just naturally helped each other out.” [Student 04]

“We helped each other out pretty naturally. It felt like we were really in it together, not just doing individual assignments.” [Student 06]

“Our group was the main reason I felt connected to the course. Without the team, it would have felt really isolating.” [Student 08, translated]

“We had one person drop out, and it threw us off, but we adjusted. That process actually brought the rest of us closer together.” [Student 02]

Students consistently described the team as their primary point of connection in the course. Team-based work was seen as central not only to completing the challenge but also to feeling anchored in the course community. While experiences varied depending on group dynamics, most students felt that sustained collaboration over weeks created a sense of shared responsibility and interpersonal familiarity. Some also mentioned tensions when roles were uneven or when one member disengaged, but even these challenges often led to stronger internal bonds among the remaining members.

Recurring themes in this category included: peer reliance, collaborative identity, group continuity, and interpersonal adaptation.

The role of the team for SoB is crucial, and the Openlab course design includes team participation for all course participants. The team context plays a crucial role in shaping stu-

dents' sense of belonging, particularly in programmes that rely on collaborative pedagogies or project-based learning. Studies have found that peer interaction, task interdependence, and supportive group dynamics can significantly influence students' emotional and academic integration (Hausmann et al., 2007; Peacock et al., 2021). When teams function well, they offer not only cognitive scaffolding but also a socio-emotional anchor, helping students situate themselves within a network of shared goals and interpersonal trust. The team accordingly became a starting point for the interviewees' explanations of SoB and the starting point of their collaboration, as well as the arena to deal with team participants dropping out. Not all responsibilities/roles in the teams were fully satisfactory. One student, a resident of Stockholm, felt a responsibility to explain the specific culture in relation to the team's assignment. Much time was, for example, spent on explaining high staff costs in Sweden, influencing hypothetical solutions to their challenge.

4.4. Stakeholder Missions (Called Challenges in the Course)

In the context of the Openlab course, a "challenge" refers to a real-world problem or task assigned by an external organisation such as the City of Stockholm. These externally defined challenges form the core of the course structure and require sustained teamwork to explore, understand, and address them. Several students described how engaging with such a task collectively helped build a shared identity and purpose within the group.

"I felt a sense of belonging in our group because we shared the experience of taking this course, of being involved in this project, of being given this task or challenge, and of collectively trying to solve it." [Student 08, translated]

"We all worked as a team on the same problem from the City [of Stockholm], and that made it feel like a real mission. It wasn't just a class assignment." [Student 12, translated]

"Knowing that this challenge came from a real organisation made us take it seriously. It gave us a shared purpose." [Student 11]

Many students described the externally defined challenge—a real-world task provided by a municipal stakeholder—as a central anchor for collaboration. The fact that the challenge was not hypothetical, but drawn from real needs in society, gave students a sense of working toward something meaningful. This created a shared focus and helped groups establish cohesion through mutual engagement with a common objective. Rather than viewing the challenge as a source of stress, most participants saw it as a unifying feature of the course structure.

Commonly cited themes in this category included: shared mission, real-world relevance, external anchoring, and group focus. Here, the student highlights how the act of facing a real-world assignment together fostered a sense of belonging. It was not merely the difficulty of the task that contributed to this experience, but the fact that it was a common, externally anchored objective that required coordination and joint problem-solving. For many, the challenge served as a unifying structure that gave meaning and continuity to their collaboration.

In several interviews, students expressed a sense of belonging that emerged through shared engagement in the project challenge. For example, one student noted: 'I felt a sense of belonging in our group because we shared the experience of taking this course, of being involved in this project, of being given this task or challenge, and of collectively trying to solve it.' This statement illustrates how the challenge itself became a **vehicle for shared experiences**, and how the experience of problem-solving together fostered inclusion and cohesion.

When working towards a common goal or challenge, the students build up the SoB in the team. The feeling can be achieved by sharing experiences and working together

to solve problems. The SoB among the team members was raised, even if the team was still determining whether or not they should fulfil the challenge givers' high expectations. Also, the assignment of projects generated SoB as one student expressed, *"Thank God we got this [challenge] and not any of the others."* [Student 08, translated]. Several students mentioned that using pass/fail grades instead of more extensive grading can encourage student collaboration. From the perspective of SoB, this is an important factor in order to have the students/teams heading for the same goals to solve the challenge.

4.5. Academic Channels

"It always feels luxurious this way, that there are so many teachers here and it really feels like an important thing, different from a regular course, and they have their own nicely designed webpages." [Student 12, translated]

"It felt more structured than other online courses. There were a lot of teachers involved, and they responded quickly when we asked questions." [Student 08, translated]

"I didn't feel the need to contact the teachers all the time because we had a strong team, but it helped to know they were there if we got stuck." [Student 04]

"Some of us had less time to socialise, so being able to check instructions or updates directly from the teachers was essential." [Student 11]

Students reflected on their interactions with teachers and the course infrastructure as part of their overall learning experience. While most participants emphasised the central role of their team, the presence of accessible and responsive teaching staff contributed to their sense of stability and support. Several students noted that teacher availability was especially important during the early stages of the course or when navigating assessment moments. Others appreciated the design of the course platform as professional and easy to follow, even if it was primarily used for one-way communication.

Themes identified in this category included: teacher responsiveness, institutional presence, academic reliability, and structural clarity. These findings align with [Martin and Bolliger \(2018\)](#), who identified instructor presence, timely responses, and structured interaction points as key components of perceived engagement in online courses.

The academic communication channels include different ways for the students to connect to the teachers and access the course material. The learning management system utilised in the course was mainly used for one-way information. More critical was the synchronous and asynchronous communication with the teachers, particularly at the beginning of the course. However, the need for quick communication between students and teachers decreased, according to the students, since tight teams were part of the course design. The team can often interpret a situation and devise a satisfactory solution without contacting a teacher. The interviewees also gave examples of older students with life situations with a lot of work and family duties who had lower preferences for group work that included socialising ingredients and expressed a higher dependency on communication channels directed to the course management.

4.6. Quick Channel

"Only for the participants. Those questions you [normally] ask during the break." [Student 11]

"We had a WhatsApp group from day one. That's where all the real talk happened." [Student 03]

"The official channels were too slow sometimes. In the chat we could ask quick things, even late at night." [Student 06]

Students strongly valued having fast, informal communication channels within their teams, separate from the official course systems. These tools, such as WhatsApp or similar apps, allowed for immediate, unfiltered interaction and helped teams build familiarity beyond scheduled meetings. The chats were used not only for logistical coordination but also for humour, encouragement, and emotional check-ins. Several participants noted that these quick channels helped compensate for the absence of spontaneous, physical “hall-way conversations” typical of on-campus experiences. Recurrent themes in this category included: immediacy, informality, peer bonding, and private space.

The lack of an informal “coffee machine chat” is often mentioned as a disadvantage in online education. This can be mitigated by simple intra-team communication channels (S. M. Ross, 2019). According to the course participants interviewed, the chat possibilities should be quick and easily accessible and separated from the formal communication channels. During the team formation phase, the teams are required to select and set up a communication channel such as WhatsApp or Slack to facilitate almost instantaneous communication within the group, not only during course hours. This way, most teams create bonds that also stretch outside of school hours and venues.

4.7. Shared Systems

“It is not the system as such that contributes to belonging, but rather the knowledge of how to use it, because I believe that if you as a class or group have a system that you work in, and you are comfortable using it, it contributes to increasing the sense of belonging.”
[Student 12, translated]

“One of our team members set up a project board, and that really helped us stay on track. It made us feel like a proper team.” [Student 06]

“If someone didn’t know how to use the tool, we would help them, so no one felt left out.”
[Student 03]

Students highlighted the importance of shared digital tools—not just their presence, but the team’s collective fluency in using them. Systems such as shared drives, messaging platforms, or project boards became central hubs for organising work and maintaining momentum. Several students noted that helping each other master these tools fostered a sense of inclusion and reduced friction. Rather than the platforms themselves fostering belonging, it was the collaborative use and mutual support around them that made a difference.

Thematic elements in this category included: collective tool use, inclusion through competence-sharing, and digital fluency as a bonding mechanism.

The students’ use of shared IT systems and that all team members have the skills to use the system effortlessly is essential to reinforce the SoB. The interviewees gave examples of how one team member keen on project management implemented a simple project management system for the team. It is relevant both for the work per se and for strengthening the team, as the participants share a skill that might not exist outside the team. No one is left outside feeling “un-belonging”. However, it is important to monitor the IT systems used in the course for signs of abuse as well as miscommunication.

4.8. Residual Observations

In addition to the seven observed categories, some observations did not fall into any of those categories but were still of interest when the study was finished. They are collected in an eighth residual category.

The wish to use information about students’ activities has also increased in line with the improved access to and use of student systems such as learning management systems and other digital meeting places. Many setups aim to decrease the time lag between

identifying a student's difficulty and the actual at-risk state, without keeping the teacher constantly online (Bañeres et al., 2020). Obtaining information early on increases the possibility of reducing the risk that the student falls behind and does not catch up, and with a later dropout as a result, a substantially higher risk in an online setting where teachers have limited opportunities to observe events unfolding or not taking place. The way to reach out to the student can be through personal contact and emails sent out automatically. Several studies have found that this kind of feedback, even automatic ones, is highly valued for the students' inspiration, particularly in asynchronous or online learning settings (Bañeres et al., 2020; Henri et al., 2021). The information from the at-risk students makes it possible to identify an ill-explained activity when additional help is needed and difficult sections in the curriculum, thereby making it possible to redesign and decrease the number of dropouts.

However, Henri et al. (2021) warn of "dishonest signals" and "stereotyping" in student-tracking information systems. A dashboard informs users about attendance and engagement but may not consider student learning diversity. An at-risk indication can also consolidate students' and teachers' doubts about possible success. Early warning systems can be based on rather complex algorithms and recommender systems that provide the students with suggested readings based on other students' ratings. However, less complicated functionality is of high value. A posting in an asynchronous discussion forum or any contribution from a student without any response can be highly demotivating for an individual student. The possibility to automatically inform the teacher or any group leader about absent feedback is a valuable part of the warning system and contributes to the sense of belonging.

Another important residual factor is to try to find a way to recognise the student participation in the online course or programme. This could be accomplished by a (non-formal) diploma or other visible recognition of participating in the online context. When clustering students' responses about SoB, Ahn and Davis (2020a) found four domains, as discussed earlier: the domain of surroundings refers to the university's geographical, environmental, and cultural contexts. Interestingly, more than half of the students mentioned one or more words related to the surroundings when contributing to the word cloud in the study. They expanded the functionality of the (campus) university beyond teaching functionality. Some responses could be traced back to the university's promotional material. Therefore, the branding of the university and students' perception of a high-status institution are relevant to the concept of SoB. However, the essential differences among the students, such as national-dependent geographies and traditions in the community, play different roles. Some participants in the Openlab course also highlighted the (international) statuses of the respective higher education institutions involved in the Openlab collaboration as an important factor.

5. Discussion

The following discussion builds on the seven thematic categories derived from student interviews. Each section examines how the reported experiences align with or extend current understanding of sense of belonging in online, project-based learning contexts. The view mentioned earlier—not desiring to make friends—was related to the interviewee's epistemological view on learning, arguing that time spent on socialising takes time from penetrating and trying to learn the content. It is not a question only about combining older and working students with young students appreciating life on campus. It is also about preferences for learning and self-regulation skills. Kuo (2010) also emphasise the role of self-regulated learning and digital self-efficacy in shaping student outcomes and engagement in online courses, reinforcing the relevance of such factors in the Openlab

setting. Such diversity has been shown to be very beneficial for enhancing creativity and innovation capacity in design thinking teams, and we will discuss the consequences of these findings in the next part. It also highlights the teams' importance and the challenge presented to them. We can assume that the finding of the importance of the team and the challenge that influenced their work in the teams is rather different compared to a traditional online, campus or blended course setting.

5.1. Building Trust

The early-stage establishment of trust within student teams can be understood as both a psychological and structural condition for collaboration, particularly in online formats where informal cues are limited. The use of a team contract in this setting goes far beyond its administrative function; it acts as a scaffold for what [Salmon et al. \(2010\)](#) call "online socialisation", i.e., the structured creation of interpersonal presence in digital environments. In this sense, the contract mediates not only expectations but also belonging, by providing an artefact around which group identity can be anchored.

The Openlab practice of pairing the contract with a group dynamics workshop reflects an intentional effort to support [Tuckman's \(1965\)](#) storming–norming transition. Rather than avoiding friction, students are equipped to interpret conflict as a developmental phase, which aligns with [Wheelan's \(1990\)](#) emphasis on the necessity of structured progression through group maturity stages. Notably, the emotional safety required for this progression, a core condition for SoB, is often contingent on pre-emptive norm-setting. In this respect, trust is not a by-product of time or familiarity, but an emergent property of early pedagogical design.

Moreover, in the context of online learning where asynchronous interaction may reduce opportunities for informal repair of missteps, codified expectations serve a preventive function. While [Hesterman \(2016\)](#) frames such contracts as technological artefacts, in this study their primary impact was social: the shared sense of accountability they fostered appeared to support student resilience during membership changes or uneven participation. This suggests that belonging, in project-based online teams, may be less about persistent presence and more about agreed commitment, a distinction with course design implications.

5.2. Personal Profiles

The role of digital personalisation in fostering social anchoring must be understood as context-sensitive rather than universally positive. In hybrid or fully online learning settings, where spontaneous hallway interactions are absent, structured personal disclosures, such as names, photos, and short biographies, help reduce initial ambiguity and establish interpersonal context. This aligns with [Salmon's \(2000\)](#) second stage of online learning: "Online Socialisation," in which learners begin to develop mutual awareness and interpersonal trust. The check-in rounds and profile features used in the Openlab course served this function by creating low-stakes opportunities for students to present themselves and to perceive others as real, present participants in the learning environment.

However, the findings also indicate variability in how these personalisation features were received. While some students valued photos and brief facts as ice-breakers, others engaged less with profiles and preferred oral introductions during live sessions. This heterogeneity underscores a recurring challenge in online and hybrid course design: balancing structured socialisation with respect for individual boundaries and preferences. Prior research (e.g., [Kear et al., 2014](#); [Kim et al., 2009](#)) similarly notes that while visible social presence promotes belonging and engagement, excessive demands for self-disclosure can backfire, in particular among students with privacy concerns or unfamiliarity with digital

platforms. As such, the effectiveness of profile features is likely moderated by both cultural context and platform literacy.

These observations suggest that personalisation strategies should be integrated not as standalone technical features but as part of a broader socialisation pedagogy. Encouraging, but not requiring, students to contribute personal details can allow them to calibrate their level of exposure. Providing opt-in mechanisms, such as optional profile fields or guided check-ins with preselected prompts, may help foster a shared baseline of presence without enforcing artificial intimacy. The broader implication is that social anchoring, when handled with flexibility, plays a formative role in shaping the trajectory of team cohesion and belonging. As [Jones and Peachey \(2005\)](#) caution, extended socialisation stages should not delay progression into academic engagement, but when timed and scaled appropriately, they serve as bridges into collaborative practice.

5.3. Teamwork

Teamwork in project-based courses like Openlab appears to function not merely as a pedagogical strategy, but as the primary social architecture through which belonging is enacted and sustained. The students' accounts underscore a relational model of SoB—it was through consistent, shared activity and mutual adaptation that a sense of connection developed, especially in the absence of traditional classroom cues. This suggests that belonging is not a static feeling established at entry, but an emergent condition shaped by collaborative continuity. The recurring mention of “natural” collaboration reflects more than interpersonal ease; it signals that well-framed team structures can internalise norms of reciprocity and responsiveness, making SoB a by-product of joint action rather than imposed structure.

At the same time, moments of disruption, such as team member dropout or uneven role distribution, acted as pressure points that tested group cohesion. Rather than undermining SoB, some of these events prompted teams to recalibrate and strengthen their internal dynamics. This supports prior work by [Scogin et al. \(2017\)](#) and [Brodie and Osowska \(2021\)](#), who argue that resilience in project teams is often a marker of mature collaboration. The Openlab model, with its extended team timelines and task interdependence, provided enough temporal and structural depth for these shifts to occur. This is a key difference from short-cycle or loosely coupled group work, where such depth may be lacking.

This dynamic also reflects the importance of group identity as an anchor point in online education. When students saw the team as their main connection to the course, it effectively became the container for both academic and emotional engagement. That centrality, however, places a burden on initial group formation, including clarity of roles and equitable distribution of tasks. Drawing on [Oakley et al. \(2004\)](#), these early conditions may determine whether teams evolve into inclusive communities or reproduce frustration and fragmentation. In Openlab, students' varying experiences with task roles—including cultural explanations or taking responsibility for specific perspectives—also reveal that belonging is relational but not symmetric. It may require some to carry interpretive or communicative labour unequally. Recognising and designing for this complexity is essential when teams are not only a means to an end, but also the space where students come to feel they belong.

5.4. Challenges

It is worth noting that “challenge” in this study refers to the externally provided course assignment, not to general difficulty or adversity. While these tasks were often complex and open-ended, their role in supporting SoB was primarily through shared engagement and collaboration, not stress or struggle. Besides being a challenge that has originated from an

actual real-life problem that the challenge giver needs help to find new innovative solutions to, the below criteria are used by Openlab as guidelines for the selection: That the challenge is a complex societal challenge that can be connected to the overall mission of Openlab; That the challenge has a scale-up potential; That the challenge targets, to some extent the global SDG goals; That the challenge is anchored within the management team of the challenge giver organisation; That the challenge fits the learning outcomes of the course.

The importance of the externally defined challenge in this course reinforces key insights from the project-based learning (PBL) literature. Real-world tasks have long been recognised as powerful motivators for student engagement and identity development, particularly when they are complex, authentic, and socially relevant (Blumenfeld et al., 1991; Bell, 2010). In the Openlab course, the challenge served not only as the object of learning but as a shared external anchor that facilitated team cohesion and goal alignment. Holm (2011) notes that projects with direct societal application tend to promote collaboration and self-regulation, while J. W. Thomas (2000) emphasises that sustained, meaningful inquiry encourages group bonding and the development of a learning community. Our findings show that students in Openlab drew much of their sense of belonging from the shared purpose inherent in tackling a stakeholder-driven challenge, a pattern that aligns closely with these earlier studies.

The role of challenge in fostering a sense of belonging may be underappreciated in prior studies focused mainly on social or emotional variables. In our study, students often framed their belonging not despite the challenge but because of it. This aligns with Hmelo-Silver's (2004) assertion that ill-structured problems promote both cognitive engagement and peer interdependence. Similarly, J. W. Thomas (2000) and Scogin et al. (2017) highlight that authentic challenges in PBL contexts demand mutual support, role negotiation, and real-time collaboration—conditions that appear to catalyse social cohesion and inclusion.

5.5. Academic Channels

The students' reflections on communication channels and teacher responsiveness suggest that academic belonging is not merely a function of peer interaction, but also of perceived institutional presence. Even when the teaching team was not frequently contacted, its availability (and the design of communication systems) contributed to the students' sense of stability and legitimacy. This aligns with Martin and Bolliger's (2018) findings that timely instructor responses and structured points of contact are key predictors of student satisfaction in online courses. In online environments, where formal authority is often less visible than in face-to-face teaching, the perceived responsiveness of academic staff plays a symbolic role: it signals that someone is paying attention, and that the course has weight.

Interestingly, the data also suggest that the need for direct teacher interaction diminishes when teams are strong and well-functioning. In such cases, teams act as first-line interpretive filters. This is in alignment with previous findings (e.g., Brodie & Osowska, 2021) that peer scaffolding can reduce dependency on formal instruction. However, this does not imply that academic channels are peripheral. Rather, they serve as infrastructural reassurance, especially for students who are time-constrained, returning to education, or less integrated into the group dynamic. These students reported higher reliance on direct guidance from the teaching team and valued asynchronous access to structured course information.

The challenge, therefore, lies in designing academic channels that are both flexible and visible. While response latency can discourage engagement, particularly in assessment phases, students' sense of belonging is shaped as much by knowing where and how to seek support as by the frequency of using those supports. This reflects a broader

pattern in SoB literature: institutional practices that project care and coherence, even when passively consumed, reinforce students' internalisation of being part of a legitimate academic endeavour. Responsive yet non-intrusive academic channels may thus act as silent anchors, particularly in online formats, where learners oscillate between self-directed study and collective engagement.

5.6. Quick Channel

In hybrid and online education, spontaneous peer interaction is often the first casualty of digital mediation. The hallway conversation, the casual reassurance during a break, or the offhand comment after class—these moments are structurally absent in most formal platforms. What emerged in this study is how students used informal, rapid-response channels (e.g., WhatsApp) to recreate these missing interactions. These spaces functioned not only as logistical lifelines but as social glue—fostering a layer of interpersonal connection that was flexible, candid, and emotionally resonant. Their immediacy enabled what might be called micro-belonging: a low-friction, high-frequency sense of being in sync with others.

Unlike institutional learning management systems (LMS), which are typically asynchronous and structured around curricular logic, these informal channels supported relational immediacy. Students did not merely use them to ask questions—they shared jokes, voiced frustrations, and coordinated everyday life. As [J. Ross \(2019\)](#) argues, the absence of casual, embodied presence in online settings can undermine students' sense of inclusion unless counterbalanced by low-threshold forms of engagement. These “quick channels” did exactly that: they allowed belonging to be enacted through rapid back-and-forth, not just formal recognition or assignment feedback.

Importantly, the students' preference for non-institutional tools also reflected a need for ownership over their communicative space. [Fehrman and Watson \(2021\)](#) note that tool choice subtly shapes discursive norms, where LMS discussions tend toward task orientation, mobile chat platforms invite socio-emotional transparency. The implication is not to replace formal systems, but to recognise that informal digital spaces are not peripheral; they are central to students' social ecology. In online education, enabling (or at least not obstructing) the creation of such informal peer channels may be one of the simplest ways to support continuous, authentic belonging beyond class hours.

5.7. Shared Systems

While technical platforms do not inherently produce belonging, they become vessels for it when shared competence and mutual support shape their use. The students' accounts suggest that collaborative fluency with digital tools—from shared drives to project boards—served not only practical coordination but also symbolic inclusion. Helping others navigate a platform or co-develop a workflow became an act of social anchoring, reducing barriers to participation and reinforcing the sense that “we are in this together.” This resonates with [Wenger's \(1998\)](#) theory of communities of practice, where shared repertoires and mutual engagement constitute the basis for identity and inclusion.

Importantly, the findings show that it was not tool choice alone that mattered, but the group's collective sense of ownership and accessibility. A tool implemented by one team member became meaningful only when others understood and adopted it. This collaborative uptake fostered a sense of co-agency, a feeling that the group was not merely following instructions but shaping its own working culture. As [Lave and Wenger \(1991\)](#) suggest, participation in shared practices is not just instrumental; it is constitutive of membership. In this context, digital fluency became less about individual mastery and

more about co-fluency, i.e., the shared ability to navigate the tools underpinning the team's collaboration.

Moreover, support around digital tools carried a quiet affective dimension. When students helped each other with unfamiliar platforms, they enacted care, patience, and mutual investment, all foundational elements of SoB. Kuo et al. (2014) affirm that digital confidence correlates with inclusion, but this study nuances that claim; what mattered was not just being confident, but being supported in becoming confident. Course design should therefore approach digital tools not only as infrastructural assets but as social artefacts, where fluency is a site of belonging, and inclusion is built through everyday assistance, not formal instruction alone.

5.8. General Discussion

The course was forced online in March 2020 due to the pandemic and stayed online for over two years. During that time, the factors above were collected and observed, and changes were made to the course accordingly. After each round of the course (each semester), the course is evaluated by a formal course evaluation consisting of 18 questions, of which five are multiple-choice, 12 are free text, and the last is an open-ended catch-all text area. The course went online in 2020 and stayed online until the spring of 2022. With a course running each semester, for the full semester, this generates some data over time as well as anchor points before and after the online period. Around 70% of all students saw definite positives with the abovementioned interventions to the learning context. The main negative point was being forced online in the first place due to the effects of the pandemic. Since the course is highly interactive, with students working in project teams of five to eight students most of the time (more than 15 h per week), it was one of the last courses at Stockholm University to have the pandemic-imposed restrictions lifted. However, to sum up the evaluations, they indicate that interventions improved the SoB in the above categories and that some of them, which were kept after the course went back to campus teaching, also improved the general quality of learning in the course.

The findings of this study are in alignment with and complement recent results obtained using the University Belonging Questionnaire (Slaten et al., 2018), particularly with regard to the structural conditions that promote or inhibit students' sense of belonging. While the present study focused on the relational alignment between students and their programme environment (such as experiencing programme structure as coherent and motivating) the UBQ introduces a parallel emphasis on institutional support and perceived competence. In both studies, perceived institutional engagement with the student experience emerged as central: when students feel supported and able to navigate the expectations placed upon them, both their motivation and their subjective sense of belonging increase. This convergence suggests that belonging is not merely an interpersonal experience but is also grounded in the clarity and consistency of educational structures.

Moreover, the UBQ's identification of competence and connectedness as key sub-dimensions of belonging aligns with this study's finding that students who perceived their programmes as well-organised and their learning as meaningful also reported higher study motivation. Although the methods differ—one relying on psychometric scale validation and the other on regression-based analysis of student survey data—the pattern is remarkably consistent. Both studies reinforce the idea that students' academic integration depends not only on peer relations or self-concept, but also on institutional design and pedagogical coherence. Taken together, the two approaches suggest a stable and generalisable link between perceived academic belonging and the motivational climate fostered by educational programmes.

The implications of this study should be understood in the context of the Openlab course design. As a project-based, challenge-driven course with external stakeholders, Openlab requires students to form interdisciplinary teams, take shared responsibility for complex real-world tasks, and interact regularly with non-academic actors. This unique combination of pedagogical elements fosters a learning environment where collaboration is not optional but essential. Many of the factors contributing to students' sense of belonging (such as team cohesion, trust, informal communication, and shared tools) are directly linked to this context. As such, our findings are most applicable to similar courses that emphasise sustained group work and societal engagement.

These findings reinforce prior research linking collaborative learning structures to increased social and emotional engagement. In particular, our results support the notion that project-based learning environments can, when carefully designed, facilitate SoB by embedding students in purposeful, team-based tasks over time. This aligns with [Scogin et al. \(2017\)](#), who found that extended peer collaboration in STEM PBL courses promoted a sense of academic community. Likewise, [Salmon et al.'s \(2010\)](#) five-stage model supports the idea that digital group formation and socialisation must be actively scaffolded, especially in asynchronous contexts. The real-world anchoring of the Openlab challenges appears to enhance the relevance of peer interaction, suggesting a synergy between project-based learning and belonging that might be underutilised in more conventional online course designs.

6. Summary

This study investigates several factors contributing to a sense of belonging (SoB) in online learning environments, specifically focusing on a recurring project-based master's course in social innovation offered by Openlab Stockholm ([Openlab, 2025](#)). The course integrates real-world societal challenges and collaboration with external stakeholders, providing a unique setting for exploring SoB. Data was collected through semi-structured interviews with eight students, enabling the identification of key themes influencing their experiences.

The results show the significance of team dynamics, trust-building, and the use of team contracts in fostering SoB. Team contracts effectively created a shared understanding of roles, expectations, and conflict resolution strategies. The importance of personal profiles, including tailored introductions and shared skill sets, emerged as another critical factor in establishing initial connections among diverse students. Moreover, shared goals—represented by tackling societal challenges—helped unite students, fostering collaboration and group cohesion. Digital tools and shared systems facilitated communication and task management, although participants' familiarity with these systems varied. Quick and informal communication channels, such as team-specific messaging apps, were also instrumental in bridging gaps in online learning. The study underscores the interplay between cognitive, emotional, and social dimensions in cultivating SoB. These results indicate the need for inclusive and innovative strategies in online education to mitigate isolation and enhance student engagement and, thereby, satisfaction.

7. Conclusions

This study demonstrates how students' sense of belonging (SoB) is shaped by the structure and cultural framing of their educational environment, particularly in online, project-based settings. Among the seven identified categories, practices such as co-created team agreements, personal introductions, and shared goal-setting emerged as especially influential. These findings echo earlier research that links belonging to structured social and pedagogical integration ([L. Thomas, 2012](#); [Kahu et al., 2020](#)). In line with the Uni-

versity Belonging Questionnaire (UBQ), the importance of perceived competence and institutional support was mirrored in students' emphasis on clarity, progression, and responsive feedback.

The study also highlights the enabling role of digital tools when used deliberately to support informal interaction and group coordination. This aligns with recent findings by [Peacock et al. \(2021\)](#) and [Deng and Tavares \(2021\)](#), who note that belonging in online environments depends less on physical proximity and more on sustained communicative presence and institutional transparency. The diversity of student preferences and digital competencies in our data supports calls for flexible, multimodal learning structures that accommodate both independent and collective engagement ([Lowenthal et al., 2020](#)).

While grounded in a specific case, the study's implications extend to broader programme and course design. Integrating cognitive, emotional, and socio-cultural dimensions into educational practice—especially in contexts where in-person interaction is partial or irregular—remains essential. Rather than relying on format alone, institutions must focus on how structure and culture are perceived by students. As [Zhao and Watterston \(2021\)](#) argue, post-pandemic higher education must address not only logistical but also relational challenges. This study offers a concrete contribution to that agenda by foregrounding the programme-level conditions under which belonging can flourish.

8. Limitations

The results are limited as the study is conducted in one type of course, albeit two instances, and the questions and interviews are influenced by both the structure of the specific course type and the teacher activities. Since one author is the executive director of Openlab Stockholm and the other is the course director and one of the teachers of the Openlab courses at the master's and PhD levels, the creation of the interview guide might have led to biases, even though external readers assessed the guide beforehand. The number of interviews was also limited, but we estimated the number to be adequate since the interviews were fairly long (50 minutes on average) and could provide sufficient information by being without precise time limits and having a semi-structured form.

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Institutional Review Board Statement: Stockholm University's guidelines based on the Ethical Review Act makes this study exempt from ethical review requirements.

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

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Appendix A. Interview Data

Main Subject	Language	Completed Course?	Programme Student	Approximate Age
Computer science	English	No	Yes	20s
Social work	English	Yes	No	40s
Curative art	English	No	Yes	30s
Economics	Swedish	Yes	Yes	30s
Public health	English	Yes	Yes	20s
Open e-government	English	Yes	Yes	20s
Sustainable technologies	English	Yes	Yes	20s
Business administration	Swedish	Yes	No	40s

Four students were from Sweden (although two of them preferred being interviewed in English), the others from Finland, Germany (2), and Portugal.

Appendix B. Coding Categories and Support in Interviews

Categories	Number
Trust within teams	1
Personal profiles	2
Teamwork	3
Stakeholder challenges	4
Academic channels	5
Quick channel	6
Shared systems	7

Categories	1	2	3	4	5	6	7
Interview A	x	x	x	x			x
Interview B	x	x	x	x			
Interview C	x		x			x	
Interview D		x	x		x	x	x
Interview E	x		x		x		
Interview F	x				x		x
Interview G	x		x			x	
Interview H	x		x	x			

Appendix C. Interview Guide

The semi-structured interview guide was developed to explore how students in a project-based master's course perceived and experienced a sense of belonging (SoB), particularly in relation to course structure, collaborative dynamics, and digital systems. The guide was used for all participants and allowed for probing follow-up questions where appropriate.

Appendix C.1. Introduction and Context

Participants were briefly informed that the study concerned students' sense of belonging in hybrid (spring 2020) and online (autumn 2020) project-based courses. It is explained that prior research has linked SoB to study success and retention, and that online and lifelong learning settings may amplify the importance of community and support mechanisms. The setting includes students with diverse backgrounds and commitments, as well as collaboration with external stakeholders.

Appendix C.2. Interview Topics and Sample Questions

Background and Prior Experience

Can you describe your earlier experiences as a student?

Have you mainly studied online or on campus?

Are you currently engaged in other courses, work, family obligations, or hobbies?

Course Motivation and Expectations

Why did you apply for the Openlab course?

Has the course met your expectations so far?

Participation and Performance

How would you describe your participation in the course?

Are you satisfied with your own and your group's performance? Why or why not?

Collaboration and Interaction

How would you describe the collaboration in your group?

What was your experience collaborating with teachers, stakeholders, peers, or administrative staff?

Sense of Belonging

How important is it for you to feel connected to other students in the course? Why?

Who or what contributed most to your sense of belonging?

Were there any activities or assignments that helped you feel more connected?

Digital Tools and Learning Environment

What role did the learning management system (LMS) play in supporting or hindering your sense of community?

Have you used any other tools or platforms (e.g., Mural, Miro, Slack)? In what ways?

How did the use of physical or digital venues (e.g., Openlab) influence your experience?

Community and Ecosystem Perspective

Based on your experience, what contributes to a sense of community in this type of course?

Do you feel that you mattered to your peers, or that they mattered to you or your group?

Notes

¹ <https://openlabsthlm.se/> (accessed on 22 January 2025).

² The course has more than ten nationalities each semester. The majority of the students do not speak Swedish.

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